



SECJ3553 ARTIFICIAL INTELLIGENCE

Section 03

ASSIGNMENT 3 INTELLIGENT AGENT (5%)

Theme: Agriculture

AI-Powered Predictive Analytics for Optimised Crop Yield and Risk Mitigation

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1.0 PEAS Model

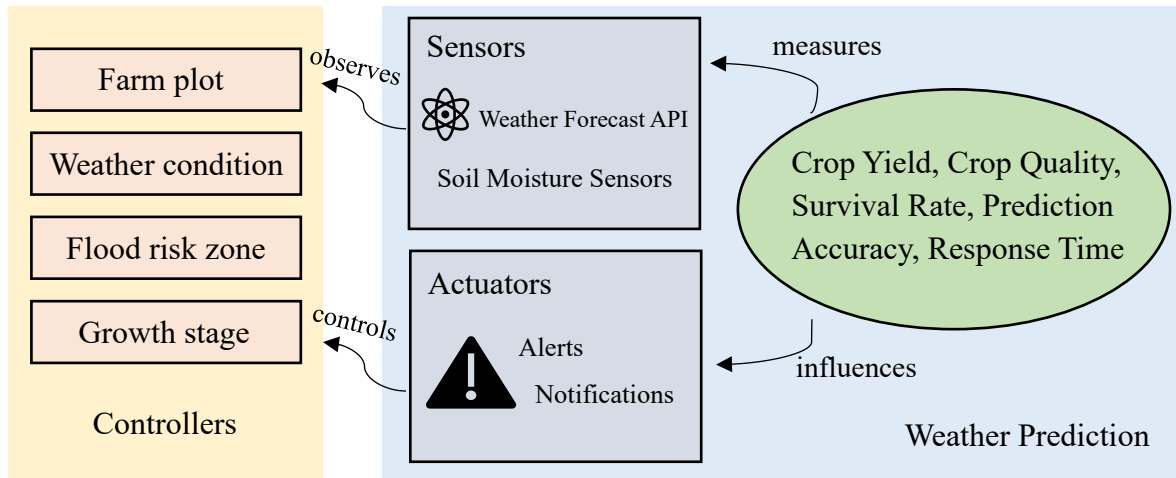
1.1 Formulation of Proposed AI Solution with PEAS Model

Property	Description
Performance	Crop Yield, Crop Quality, Survival Rate, Prediction Accuracy, Response Time
Environment	Farm Plot, Weather Conditions, Crop Growth Stage, Flood Risk Zones
Actuators	Alerts, Notifications, Yield Prediction Graph, Harvest Recommendation Interface
Sensors	Weather Forecast API, Soil Moisture Sensors, Crop Age Calculator, Flood Warning API

1.2 Explanation of Properties

Property	Explanation
P: Performance measure	The primary measure of success for the agent is the maximisation of crop yield and the minimisation of crop loss.
E: Environment	The agent is working in an agricultural environment.
A: Actuators/Effectors	The actuator is a mechanism used by the agent to affect the environment or user.
S: Sensors	The sensor is the input device used to sense the environment.

2.0 RELATIONS BETWEEN EACH PROPERTY



3.0 EXPLANATION IN PROOF OF CONCEPT

3.1 Performance

The performance of our system is evaluated by maximising the accuracy of crop yield and the prediction. Our system will be using real-time data from soil moisture sensors and the weather APIs, it will estimate the optimal planting and harvesting period of time. Also, by integrating machine learning algorithms, it will make sure our system has high accuracy of yield forecasting. It will consequently help the farmers minimise crop losses due to sudden weather changes. In our POC, the system will visualise the performance indicators through the expected yield chart on the dashboard. When the system successfully identifies the high-risk events like floods, it will show the notification of emergency harvest in time. High performance can be achieved to ensure that the crop survival rate is maintained above 90%. Besides, the system will compare the accuracy of the weather forecast with the actual weather situation. Higher accuracy is directly related to the better decision making for harvesting and irrigation and eventually improving the profitability of the farmers.

3.2 Environment

The AI agent operates in a real agricultural environment that is dynamic and unpredictable. This environment consists of farm plots, weather conditions and crop development stages as well as flood prone areas. As the weather and environmental conditions always changing, the AI agent must update immediately its understanding of the environment. The system is

designed to adapt to these changes by constantly analysing new data and adjusting its recommendations accordingly.

3.3 Actuators

The actuators in our AI agriculture system are alerts, notifications, yield prediction graph, harvest recommendation interface. The alerts and notification are there to inform farmers during critical conditions that require immediate attention. For example, during high floods, an alert will be sent to the user through the interface of the system. The farmers will also receive notifications regarding moisture levels of crops, and weather predictions that could damage the crops like heavy rainfall. This will prompt the farmers to take preventive actions to ensure that there are no crop losses. The yield prediction graph are charts that show the expected yield trend under present conditions. Visualization from the yield prediction graph can help farmers do more informed decisions and increase crop performance. The harvest recommendation interface recommends specific harvest windows based on all variables that could affect harvest. If the crops are matured and the weather is good, the interface will display a recommended harvest period.

3.4 Sensors

The sensors in our AI-powered agriculture system act as the primary data ingestion layer, consisting of soil moisture sensors and weather Forecast APIs. Firstly, soil moisture sensors detect the moisture in the soil, then provide immediate feedback on the ground conditions. This will allow the agent to detect "Humidity Low" states on the soil to ensure the humidity of soils is maintained. Besides that, Weather APIs act as long-range sensors to handle external environmental factors. Unlike local physical sensors, these APIs provide forecast data and rainfall probabilities to detect "Flood Risk" or "Drought" very well, serving as the critical input for the agent's safety logic such as "Suggest Early Harvest". Overall, these sensor inputs are the raw data that the agent's logic requires to evaluate the current state and determine if safety thresholds like safe planting windows are met.

4.0 BEHAVIOURS OF AGENT TO ACHIEVE GOAL

The project is focusing on the development of a unique AI predictive analysis system to optimise crop yields and mitigate risks and works as an intelligent agent with specific goals. The core goal is to help farmers determine the best planting and harvesting time, then maximise the crop yields whilst minimising losses caused by extreme weather conditions. The system with multiple modules also combines functions such as flood risk prediction and yield estimation.

Our system operates as a dynamic intelligent agent. It interacts with the agricultural environment through the control panel which uses multiple sensors to collect real-time data including weather forecast, soil moisture levels and crop growth stages. During the initialisation process, these sensors will collect information about the current state of the farm and take into account factors such as recent rainfall and ripe crop days.

After that, the intelligent agent will use the input data to formulate the most effective action plan for users. For instance, when the agent detects that the soil moisture is at a dangerous low level but it is predicted that it will rain heavily within 24 hours, the system prioritises the waiting strategy rather than immediately irrigating to prevent wasting. On the other hand, if it detects the risk of flooding next week, AI will trigger the emergency harvest alert.

In addition, this intelligent agent can monitor weather conditions in real time. By using real-time API data, it continuously assesses the effectiveness of the current recommendation. It provides users with quick updates to ensure that every decision is predicted. The system not only guides users to achieve the best harvest time but also optimises the overall strategy by recommending the most effective maintenance actions such as irrigation schedules whilst predicting future risks such as sudden droughts.

The feedback loop of the system involves users providing actual harvest data so that the intelligent agent can continuously improve the output prediction model. When the farmers encounter sudden environment changes such as droughts or flood, the intelligent agent provide real-time support. It may suggest alternative mitigation strategies or adjusting the expected yield curve to make sure a more stable planting. The system will improve the farmer satisfaction by reducing potential risks through predictive analysis.